

Anomaly Target Detection in Hyperspectral Images

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Detection Report

File Analyzed	abu-airport-3.mat
Report Generated	2026-06-26 00:23:01
Image Dimensions	100 × 100 pixels
Spectral Bands	205

Anomaly Detection Statistics

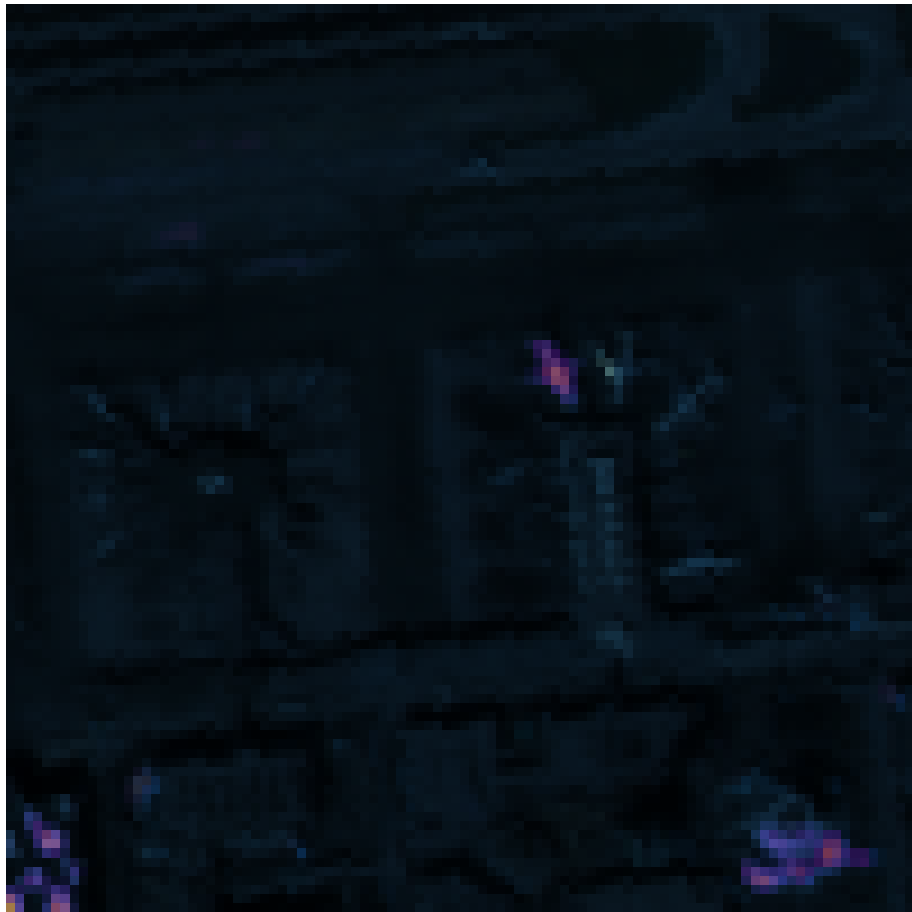
Metric	Value
Total Pixels	10,000
Anomaly Pixels Detected	163
Anomaly Coverage	1.63 %
Anomaly Score Min	0.0
Anomaly Score Max	0.731292
Anomaly Score Mean	0.005811
Anomaly Score Std Dev	0.032112

Classification Metrics

Metric	Value
AUC-ROC	0.8223
Precision	0.0396
Recall (TPR)	0.9176
F1 Score	0.0759
True Positives	156
False Positives	3785
True Negatives	6045
False Negatives	14

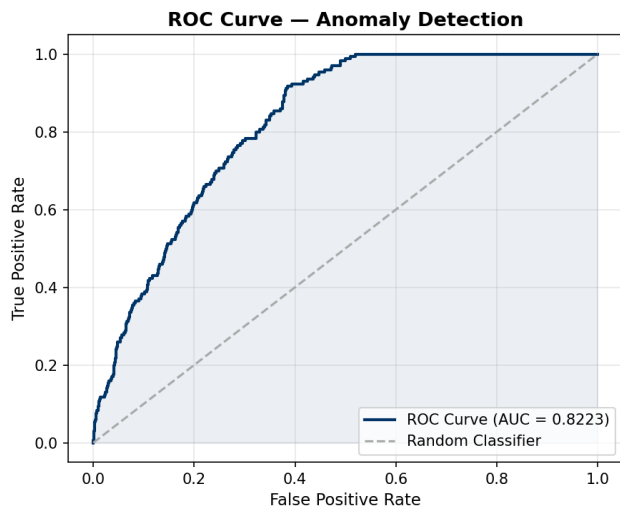
Anomaly Detection Output

Detected anomaly regions overlaid on the hyperspectral RGB composite. Bright highlighted regions indicate detected anomalies.



ROC Curve & Confusion Matrix

The ROC curve shows the trade-off between True Positive Rate and False Positive Rate across all thresholds. The confusion matrix is computed at the optimal threshold (Youden's J statistic).



Confusion Matrix

True label	Predicted label	
	Background	Anomaly
Background	6045	3785
Anomaly	14	156

AUC = 0.8223 | Threshold selected by Youden's J index

Pipeline Methodology

- 1. Preprocessing:** Hyperspectral cube normalized and resampled to 200 spectral bands.
- 2. PCA + RF + SVM:** PCA reduces to 30 components. Random Forest and SVM produce probability/decision scores as ensemble features.
- 3. LRSMD:** Low Rank and Sparse Matrix Decomposition separates background (low-rank) from anomalies (sparse) via SVD.
- 4. Autoencoder:** Deep autoencoder reconstructs background. Pixels with high reconstruction error are flagged as anomalies.